

Kapilavai Abhinaya

JAVA

PROFESSIONAL SUMMARY

- **Overview:** Full Stack **Java Developer** with over 10+ years of professional experience, adept at all **phases of software development**.
- Expertise in all phases of Software Development Life Cycle process (**SDLC**), Agile software Development, Scrum Methodology, and Test-Driven Development.
- Led Agile teams using **Scrum** and **Kanban methodologies**, managing version control with **Git**, **Bitbucket**, and **GitLab**, resulting in increased team productivity and faster release cycles.
- **UI Development:** Developed and maintained highly responsive user interfaces for financial and health applications using **Angular** (16,8, 5, AngularJS), and **ReactJS**, ensuring seamless user experiences across multiple devices and platforms.
- **Back-End Development:** Migrated a legacy monolithic application to a **microservices architecture** using **Spring Boot 3** and **Spring Cloud**, improving system scalability and resilience.
- Developed **RESTful APIs** using **Python (Django, Flask)** and **Node.js (Express.js)** for a high-traffic financial platform, ensuring secure and efficient transaction processing with **Spring Security**, **OAuth2**, and **JWT**.
- Optimized backend processing by implementing advanced **Java 8 features** such as Streams and Lambda expressions, reducing CPU usage by 20% during peak transaction periods.
- **Cloud Development:** Experience in creating **innovative** and **scalable cloud solutions** using **AWS**, integrating services such as AWS Lambda, S3, and DynamoDB to address complex requirements and drive technological advancements.
- Extensive experience integrating **Azure services** into enterprise systems and developing serverless applications. And, proficient in **Azure Functions**, Event Grid, and API Management, ensuring data protection and business continuity.
- Familiarity with **PostgreSQL** on **GCP**, **AWS RDS**, and **Amazon Aurora**, optimizing database performance and ensuring data integrity.
- **Databases:** Designed and optimized relational databases (**PostgreSQL**, **MySQL**, **Oracle**) for OptumRx, improving query performance by 40% and ensuring data integrity across all transactions.
- Managed NoSQL databases such as **MongoDB**, **Cassandra**, and **DynamoDB** to handle large-scale data storage for a Big Data project, reducing data retrieval time.
- Implemented caching strategies using **Redis** and **Memcached**, resulting in reduced server response times for high-traffic web applications.
- **API Development:** Defined and built robust **REST** and **RPC style APIs**, facilitating secure and efficient communication across distributed systems.
- Fluency in **REST Concepts** and **HTTP**, ensuring the development of efficient and scalable APIs for high-traffic platforms.
- Implemented **OAuth2**, **JWT**, and **Single Sign-On (SSO)** for a distributed microservices architecture, enhancing security and reducing unauthorized access.
- Architected and deployed **Service Bus Network (SBN)** solutions, enabling efficient and secure communication across distributed systems.
- **Testing:** Experienced in designing and executing automated test strategies for Java-based applications and APIs. Proficient in leveraging tools like **JUnit**, **TestNG**, and **Postman** to deliver robust and scalable testing solutions that drive software quality and accelerate release cycles.
- Expert in **Test-Driven Development (TDD)** with extensive experience using **Mockito** to mock external dependencies, enabling the development of clean, testable, and maintainable code.
- **Message Queuing Systems:** Implemented **Apache Kafka** for real-time data integration, supporting seamless communication between microservices and improving system reliability by 40%.
- Experienced in implementing **IBM MQ** to ensure reliable and **secure messaging** between enterprise applications, leveraging advanced features for transactional and event-driven communication.

TECHNICAL SKILLS

Programming Languages: Java (Core Java, Java 8,11,17), JavaScript (ES6+), TypeScript, Python, SQL, PL/SQL.

Web Technologies: HTML5, CSS3, jQuery, AJAX, Bootstrap, DOM, XHTML, Material-UI.

Front-End Frameworks: Angular (8, 5, AngularJS), ReactJS, Redux, React Router, Vue, Svelte

Back-End Frameworks: Spring Boot, Spring Cloud, Spring MVC, Spring Security, Spring Data JPA, Hibernate ORM, Node.js, Express.js, Django.

Microservices & API Development: RESTful APIs, SOAP, gRPC, GraphQL, Swagger.

Cloud Platforms: AWS (EC2, S3, Lambda, ECS, API Gateway, IBM WebSphere, RDS, IAM), Azure (Blob Storage, Virtual Machines, Functions, API Management), Google Cloud Platform (GCP).

Containerization & Orchestration: Docker, Kubernetes, Helm.

CI/CD & Build Tools: Jenkins, GitHub Actions, Maven, Gradle, Gulp, NPM.

Database Management: MySQL, PostgreSQL, MongoDB, Cassandra, Oracle, Redis.

Logging & Monitoring: Log4j, JProfiler, New Relic, Prometheus, Grafana, ELK Stack (Elasticsearch, Logstash, Kibana), Kibana.

Messaging & Streaming: Apache Kafka, RabbitMQ, Autosys, AMQP.

Agile & DevOps: Agile methodologies (Scrum), JIRA, Git, Bitbucket, GitLab, CI/CD.

Testing & Debugging: Jasmine, Karma, Protractor, JUnit, Mockito, Cucumber, Selenium WebDriver.

Other Tools & Technologies: Eclipse IDE, IntelliJ IDEA, Visual Studio Code, Infrastructure as Code (IaC) with Terraform.

PROFESSIONAL EXPERIENCE

Costco: July 2023 - Current (Location: Issaquah, WA)

Senior Full Stack Developer

- Developed and maintained highly responsive GUIs for applications using **React** JavaScript, **Bootstrap**, **HTML5**, **CSS3**, and **AJAX**. Ensured cross-browser compatibility and a seamless user experience across various financial platforms.
- leveraging **React** for dynamic user interactions and enhancing user experience.
- Leveraged **React's** set of **UI components** and themes to deliver responsive, user-friendly interfaces, resulting in a streamlined user experience and increased application usability.
- Developed robust integrations between frontend and backend systems, including **Spring Boot** and **RESTful** APIs, to support efficient data processing and real-time functionality.
- Migrated Monolithic legacy provider's portal to Microservices architecture using **Spring Boot 3** & **Spring Cloud**. Implemented RESTful APIs with **Spring MVC** for seamless integration between Optum financial and Optum pay.
- Utilized **Spring Boot 3** for microservices development, integrating **Spring Data JPA** and **Spring Security** with **OAuth2** to secure sensitive mortgage transaction data and ensure compliance with industry regulations.
- Designed and implemented serverless workflows using **Google Cloud Platform (GCP)** and **API Gateway** for real-time Health assessments, immunization services, and wellness programs. and improving scalability.
- Deployed and managed **Docker** containers with **Google Kubernetes Engine (GKE)**, enabling scalable infrastructure for high-volume applications.
- Built CI/CD pipelines with **GitHub Actions** and **Jenkins**, automating build, test, and deployment processes. Automated infrastructure provisioning using **Terraform**, optimizing cloud resources on GCP.
- Deployed and managed scalable applications on **AWS EC2** instances, utilizing Elastic Load Balancing (ELB) to distribute incoming traffic and ensure high availability across multiple instances.
- Designed and managed cost-effective storage **solutions** using **Amazon S3** for storing and retrieving large amounts of unstructured data, leveraging S3's scalability and durability.
- Designed and optimized **PostgreSQL** databases for mortgage applications, leveraging **Azure SQL Database** for high availability and fault tolerance. Implemented **ETL** processes to synchronize data between **PostgreSQL** and legacy systems.
- Developed and executed unit and integration test cases using **JUnit**, **Mockito**, and **Vaadin TestBench**. Employed **TDD** practices, ensuring robust Claims management, billing and coding.
- Configured **Kibana** alerts and **Grafana** dashboards to monitor key metrics and detect performance bottlenecks in **Kafka** clusters used for real-time mortgage application processing.
- Developed high-performance **REST** and **gRPC** APIs for Online therapy sessions, addiction treatment programs, and mental health assessments, ensuring fast and secure communication between microservices.
- Implemented advanced caching strategies using **Redis** and **Memcached** to enhance performance and reduce server response times for high-traffic mortgage applications.

- Leveraged **Apache Kafka** for real-time data integration, supporting seamless communication between microservices and improving system reliability by 40% for real-time mortgage data processing.
- Developed and deployed **GraphQL** APIs for flexible and efficient querying of Optum Labs -related data, improving front-end data fetching and reducing over-fetching.
- Implemented **Prometheus** for detailed metrics collection and analysis of microservices performance, complementing existing **Kibana** and **Grafana** monitoring setups.
- Utilized **ELK Stack** (Elasticsearch, Logstash, and Kibana) for comprehensive log management and analysis, enhancing visibility into application behavior and troubleshooting capabilities.
- Developed **ETL** pipelines to synchronize OptumIQ data between **MongoDB**, **PostgreSQL**, and **Oracle DB**, ensuring data consistency and integrity across multiple databases.
- Established and maintained effective incident management practices, including real-time alerting and automated remediation strategies to ensure high availability and performance of OptumIQ applications.

Environment: Java 17, Kotlin, Spring Boot 3, Spring Cloud, Spring MVC, Spring Security, OAuth2, Vaadin 21+, Google Cloud Platform (GCP), Google Cloud SQL, PostgreSQL, Docker, GKE, GitHub Actions, Jenkins, Terraform, Kafka, JUnit, Mockito, Swagger, Kibana, Grafana, Redis, Memcached, GraphQL, Prometheus, ELK Stack, Apache Solr, MongoDB, Oracle DB.

Mastercard: September 2020 - June 2023 (Location: Purchase, NY)

Senior Full Stack Developer

- **Front End Development:** Developed dynamic and responsive user interfaces using **React.js**, ensuring cross-browser compatibility and a seamless user experience for Morgan Stanley's trading platform.
- Implemented complex UI components using **React** Material and Bootstrap, enhancing the platform's visual appeal and user interaction with real-time financial data.
- Optimized front-end performance by leveraging **lazy loading**, code splitting, and **caching strategies**, resulting in a 30% reduction in page load times and improved application responsiveness.
- **Back-End Development:** Designed and developed **RESTful APIs** using **Spring Boot** to facilitate seamless communication between the front-end and back-end systems, ensuring secure and efficient data exchange.
- Integrated **GraphQL APIs** for data querying, enabling the **client to fetch** specific data with reduced network overhead, enhancing the overall application performance.
- Developed microservices architecture using **Spring Cloud**, breaking down the monolithic application into smaller, manageable services, leading to improved scalability and maintainability.
- Developed custom authentication and authorization mechanisms using **Spring Security**, implementing **OAuth 2.0** and JWT tokens to secure sensitive financial data.
- Integrated **Apache Kafka** as a message broker to handle real-time data streaming and processing, ensuring high availability and fault tolerance in the trading platform.
- **Database Management:** Implemented **Hibernate ORM** for seamless interaction with **relational databases**, ensuring efficient data persistence and retrieval for the trading platform's backend.
- Optimized **SQL queries** and performed **database indexing** to enhance the application's data retrieval speed, reducing query execution time by 25%.
- Developed and optimized complex SQL queries and stored procedures in **Oracle** and **PostgreSQL** to enhance the performance of critical financial applications, reducing query execution time by 30%.
- Utilized advanced PL/SQL techniques to develop automated processes for data extraction, transformation, and loading (**ETL**) in **Oracle** and **PostgreSQL**, streamlining data flow between multiple financial systems.
- Configured and managed **Redis** for caching frequently accessed data, reducing database load and improving application performance by up to 40%.
- **Testing:** Automated unit testing using **JUnit** and **Mockito**, achieving 90% code coverage and ensuring high-quality code delivery with reduced defects.
- Version-controlled the codebase using **Git**, collaborating with cross-functional teams via **GitHub** for efficient code reviews and feature integration.
- Containerized microservices using **Docker**, ensuring consistent environments across development, testing, and production, reducing deployment issues.
- Orchestrated containers using **Kubernetes**, managing the deployment, scaling, and monitoring of microservices, ensuring high availability and resilience.
- Implemented load balancing and scaling strategies using **Nginx**, ensuring the application can handle increased traffic without performance degradation.
- **Cloud Computing:** Designed and deployed highly available and fault-tolerant applications on **AWS** using **EC2**, Auto Scaling, and **Elastic Load Balancing (ELB)**, ensuring optimal performance and scalability during peak trading hours.

- Leveraged **AWS Lambda** and **API Gateway** to build serverless architectures, reducing operational overhead and improving the efficiency of financial data processing tasks by 35%.
- Configured and managed **AWS RDS** for PostgreSQL and Oracle databases, optimizing performance through automated backups, multi-AZ deployments, and read replicas, ensuring minimal downtime and data consistency.

Environment: ReactJS, NPM, Gulp, React Router, Redux, Node.js, Express.js, MySQL, MongoDB, Core Java, Stream API, Spring Boot, Hibernate ORM, Spring Security, OAuth2, JWT, Spring MVC, GraphQL, Flask, Django, MongoDB, MySQL, PostgreSQL, SQL, Cassandra, Oracle WebLogic, AWS, Docker, Kubernetes, Azure Functions, API Management, GitHub Actions, Jenkins, GDPR, Terraform, RabbitMQ, Kafka, Prometheus, ELK Stack (Kibana), Jasmine, Mockito, JUnit, Maven, Swagger, Redis caching, JProfile, Log4j, Git, Bitbucket.

Johnson & Johnson: November 2017 - August 2020 (Location: New Brunswick, NJ)

Fullstack Java Developer

Front-End: Developed and designed responsive web interfaces using **HTML5**, **CSS3**, **JavaScript** (ES6), **jQuery**, **Angular 8**, and **Bootstrap** for cross-browser compatibility and seamless user experiences.

- Built Components utilizing **Angular 8**, focusing on Angular directives, lazy loading, and Dynamic loading in Angular routers. Utilized TypeScript for scalable and maintainable code.
- Developed prototypes and project layouts using **Angular 8**, **jQuery**, **HTML5**, and **CSS3**, ensuring modular and reusable components.
- **Back-End:** Integrated Microservices into existing system architecture. Developed **Microservices** business components and **RESTful** service endpoints using **Spring Boot** to enhance scalability and performance.
- Developed multiple enterprise applications using **Spring Boot**, **Spring Data JPA**, **Spring MVC**, **Spring IOC**, and **Spring AOP**, ensuring robustness and maintainability.
- Implemented **Spring IOC** for **Dependency Injection** and **Spring AOP** for logging, ensuring loose coupling and cross-cutting concerns management.
- Implemented **SOAP based JAX-WS** Web services interfaces for integrating multiple systems, promoting code re-usability, and tested performance using JMeter.
- **Databases:** Designed and **optimiz**
- **ed SQL** databases using advanced relational database concepts, ensuring data integrity, normalization, and efficient query execution for high-volume healthcare applications.
- Developed complex queries, stored procedures, and functions in SQL Server to streamline data retrieval and reporting, improving system performance and scalability in critical financial projects.
- Integrated MongoDB as a NoSQL solution to handle unstructured data, leveraging its flexibility for scalable data storage in a cloud-native microservices architecture.
- Implemented data models in **Cassandra** to support distributed systems, ensuring low-latency access and high availability across geographically dispersed cloud environments.
- **Testing:** Led the design and execution of automated unit tests using **JUnit** and **Mockito** for Java-based applications, achieving 90% code coverage and significantly reducing production bugs.
- Developed integration test suites using **Selenium** to validate cross-browser compatibility and functionality for web applications, enhancing user experience.
- **Cloud Computing:** Designed and deployed microservices-based architecture on **AWS**, leveraging services such as **EC2**, **S3**, **Lambda**, and **RDS** to deliver scalable and cost-effective cloud solutions.
- Managed multi-cloud deployments using **AWS**, **GCP**, and **Azure**, ensuring seamless integration and communication between cloud platforms for high-availability applications.
- Implemented serverless solutions using **AWS Lambda** and **API Gateway**, significantly reducing operational overhead and improving system scalability.

Environment: jQuery, Angular 8, Bootstrap, TypeScript, Spring Boot, Microservices, Spring Data JPA, Spring MVC, Spring IOC, Spring AOP, Spring Security, OAuth2, Python, Django, jQuery UI, SOAP, JAX-WS, JMeter, RESTful web services, XML, JSON, Jersey, MongoDB, CRUD operations, JMS server, WebSphere Server, Continuous Integration (CI), Continuous Development (CD), Jenkins, Maven, GitHub, JUnit, Kubernetes, GCP Cloud SQL, PostgreSQL, Google Cloud Platform (GCP), Google Cloud Storage, Compute Engine, Cloud SQL, GitLab, Pandas, Django, Matplotlib, Seaborn.

Verizon: December 2015 - October 2017 (Location: Basking Ridge, NJ)

Full stack Developer

- Developed web applications in accordance with functional requirements, ensuring adherence to specifications and timely delivery.
- Implemented client-side validation using JavaScript to enhance user input handling and form validation. Designed and developed user interfaces using HTML, JSP, and jQuery for efficient and interactive web screens.
- Utilized Apache **Struts** for MVC architecture to develop web-based components, ensuring separation of concerns and maintainability. Implemented server-side modules using Java EE, Spring Framework, and Hibernate, integrating with MySQL for data persistence.
- Configured and deployed applications on Apache Tomcat and **WebLogic Application Server**, ensuring robust deployment and scalability. Implemented transactional logic in the Data Access Layer, ensuring data integrity and consistency.
- Performed unit testing using **JUnit** to validate application functionality and ensure code quality. Utilized Log4J for logging errors, info, and debugging messages, facilitating effective troubleshooting and maintenance.
- Developed transaction-oriented logic to handle complex data interactions and ensure database integrity using Spring Data **JPA and Hibernate**.
- Integrated security measures using Spring Security to manage authentication and authorization within applications.
- Utilized **Amazon Web Services (AWS)** for cloud-based solutions, including EC2 for scalable computing resources and S3 for storage solutions.

Environment: Java 7, Java EE, JSP, Servlets, Spring Framework, Spring Data JPA, Hibernate, Apache Struts, MySQL, Apache Tomcat, WebLogic Application Server, RESTful and SOAP Web Services, HTML, JavaScript, jQuery, JUnit, Log4J, SVN, PL/SQL, UML, Unix, AWS (EC2, S3).

Tally Solutions: May 2014 - October 2015 (Location: Bangalore, India)

Java Developer

- Bolstered to the development of **Java applications** by writing, testing, and **debugging** code under the supervision of senior developers. Implemented core functionalities and integrated features into existing applications, ensuring adherence to coding standards and project requirements.
- Assisted in the development of both **client-side** and **server-side** components, **participating** in the entire software development lifecycle. Identified and resolved software bugs and issues reported by users and detected during testing, ensuring application stability and performance.
- Assisted in the development and optimization of SQL queries and Java Database Connectivity (JDBC) operations to facilitate data retrieval and manipulation. Worked with database administrators to ensure seamless integration between Java applications and backend databases, maintaining data integrity.
- Collaborated with designers, testers, and project managers to ensure alignment with project goals and deliver high-quality software solutions. Contributed to peer code reviews, offering constructive feedback and learning from the review process to enhance coding skills.

Environment: Spring MVC, JDBC, Hibernate, JSP, HTML, CSS, DB2, MySQL, PostgreSQL, Jira, CVS, Junit

Education:KL University

Bachelor of Technology in Computer Science

(2011-2014)